

Luiz Takahashi

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EDUCATION

Washington State University

B.S. in Computer Science, GPA: 3.82/4.0

Pullman, WA

Expected May 2027

President's Honor Roll (4 semesters). Coursework: Machine Learning, Data Mining, Algorithms, Linear Algebra.

RESEARCH EXPERIENCE

Independent Researcher, Computational Biology (single-cell / neural organoids)

2025 – Present

Self-directed; first-author preprint in submission (arXiv q-bio.QM).

- Conducted a single-author study of the off-target compartment of human neural organoids: mapped a 1.77M-cell organoid atlas onto a harmonized 2.6M-cell fetal-brain reference (scVI / scANVI / scArches), defining calibrated off-manifold and label-entropy scores that resolve a six-state taxonomy of off-target cells.
- Established an *identifiability boundary*: proved a standard contrastive disentanglement model cannot separate lineage from in-vitro state (an adversary recovers cell source at >0.998 accuracy regardless of latent capacity), then recovered confound-invariant geometry via sliced-Wasserstein cancellation, validated with a confound-controlled cross-atlas test.

Undergraduate Research Assistant, CASAS Smart Home Lab & Neuropsychology and Aging Lab

Aug 2025 – Present

Washington State University. Advisors: Dr. Diane J. Cook & Dr. Maureen Schmitter-Edgecombe. Funded by NIH/NIA.

- Developed a room-aware Kalman filter for indoor localization from sparse smart-home sensors monitoring 35 older adults (ages 60–90+, 4 days to 22 months); reduced localization error by 18% with 92% statistical consistency (vs. 65% baseline) across 2M+ sensor events.
- Found cognitive accuracy significantly associated with behavioral variety ($r = -0.43$, $p = 0.015$), driven almost entirely by MCI participants ($r = -0.93$, $p = 0.002$); showed the effect dissolves under a noise-multiplied outcome, isolating where the signal lives.
- Built a LASSO classifier (healthy vs. impaired, AUC = 0.78) identifying spatial entropy ($\beta = -0.42$) and night-activity fraction ($\beta = +0.38$) as clinically interpretable biomarkers consistent with sundowning.

Independent Researcher (Columbia University / Wen Lab collaboration), Multi-View Interaction Hierarchy

2025 – Present

- Built an analysis pipeline from raw NHANES data ($\sim 3,000$ adults, 24 biomarkers, 8 organ systems); identified a “fog zone” (fasting glucose 100–125 mg/dL) where prediction fails, and proved mathematically why single-view predictions are blind to interaction terms.
- Proved a compression-order tradeoff theorem via Hoeffding functional decomposition; the methods are now applied by Columbia's Wen Lab to UK Biobank protein-aging data for Alzheimer's and mortality prediction.

Research Assistant, Center for Artificial Intelligence (C4AI), University of São Paulo

May – Aug 2025

HarpIA Project. Advisor: Dr. Fabio Cozman.

- Developed SQL Component Match, a semantic natural-language-to-SQL evaluation metric using AST analysis ($\sim 4,000$ lines of Python): 95% semantic accuracy vs. a 70% Spider baseline with $O(n^{1.08})$ scalability; built a 5,686-line test suite (234 methods) with multi-stage normalization, a DNF circuit breaker, and Unicode homograph detection.

PUBLICATIONS & PRESENTATIONS

- L. Takahashi.** *NullState: a reference-anchored framework for detecting, diagnosing, and quantifying the off-target compartment of human neural organoids.* In submission (arXiv q-bio.QM), 2026. (Sole author.)
- L. Takahashi, J. Wen.** Multi-view interaction hierarchy for health screening. *In preparation* (target: AISTATS 2027).
- L. Takahashi, D. J. Cook, M. Schmitter-Edgecombe.** Day-similarity structure in older adults. *In preparation* (target: journal).
- Poster:** L. Takahashi, D. J. Cook, M. Schmitter-Edgecombe. Room-Aware Behavioral Profiling for Cognitive Health Monitoring. *International Society for Gerontechnology (ISG) 2026, Vancouver, Canada.*

TEACHING

Teaching Assistant, Washington State University

2025 – Present

- Program Design & Development (C/C++):** lead a weekly 3-hour lab for 15+ students; mentor on data structures, algorithms, and debugging during office hours.
- Data Mining:** sole teaching assistant for the course; graded assignments and exams for 50+ students.

TECHNICAL SKILLS

Languages: Python (advanced), C (proficient), C++, SQL, C#, Bash **ML/AI:** PyTorch, scikit-learn, scvi-tools/Scanpy, Kalman filtering, LASSO/Ridge, KNN, neural networks

Data: Pandas, NumPy, SciPy, Matplotlib, Seaborn, sqlglot, time-series analysis, synthetic data generation **Tools:** Git, Docker, Linux, L^AT_EX, Jupyter